

Shopify Stock Analysis - Week 4

Power BI Exploratory Data Analysis | Tool: Microsoft Power BI Desktop | Dataset: Shopify_Stock with Dim_Date

1. Project Overview

This week I continued the Shopify stock dashboard from Week 3. I added moving averages, a crossover signal, a price chart and a volume chart with date controls, a metric selector and a tooltip page. I kept the Week 3 KPI cards and date range slicer and built the new visuals on top of them instead of starting again.

Tasks for this week:

- Create 20-day and 50-day SMA measures using DATESINPERIOD
- Create a crossover signal that shows Bullish or Bearish by comparing the 20 SMA and 50 SMA
- Make a price chart with Close Price, 20 SMA and 50 SMA on a continuous date axis with a zoom slider
- Make a volume chart with the 20-day SMA and highlight volume spikes
- Add a Between date range slicer and a Relative Date slicer
- Add a Metric Selector for Open, High, Low and Close using a field parameter
- Create a tooltip page called Stock_Tooltip

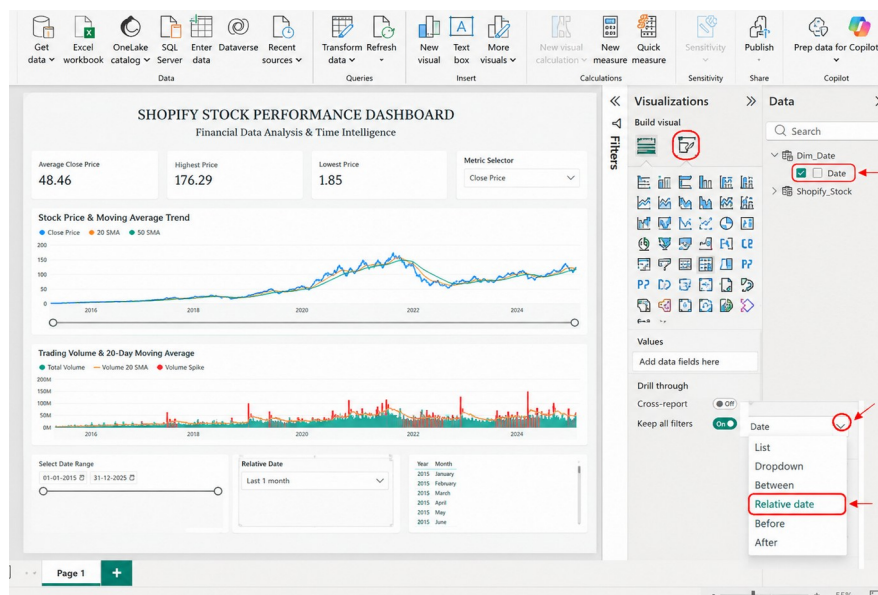


Figure 1: Power BI working view with the price chart, volume chart, metric selector and date controls

2. DAX Measures

The moving averages use DATESINPERIOD to take the last 20 or 50 days up to the current date and average the values in that window. This is the 20-day volume average:

```
Volume 20 SMA =  
VAR CurrentDate = MAX(Dim_Date[Date])  
VAR PeriodDays = 20  
VAR DateInPeriod =  
    DATESINPERIOD(Dim_Date[Date], CurrentDate, -PeriodDays, DAY)  
RETURN  
    CALCULATE(AVERAGE(Shopify_Stock[Volume]), DateInPeriod)
```

The 50-day version is the same, with PeriodDays set to 50:

```
Volume 50 SMA =  
VAR CurrentDate = MAX(Dim_Date[Date])  
VAR PeriodDays = 50  
VAR DateInPeriod =  
    DATESINPERIOD(Dim_Date[Date], CurrentDate, -PeriodDays, DAY)  
RETURN  
    CALCULATE(AVERAGE(Shopify_Stock[Volume]), DateInPeriod)
```

For the crossover signal, I compare the two averages. If either one is blank it returns blank, otherwise it shows Bullish when the 20 SMA is above the 50 SMA and Bearish when it is below. The measure names in the formula have to match the measures in the model exactly.

```
MA Crossover Signal =  
IF(  
    ISBLANK([20 SMA]) || ISBLANK([50 SMA]),  
    BLANK(),  
    IF([20 SMA] > [50 SMA], "Bullish (20 > 50)", "Bearish (20 < 50)")  
)
```

3. Price Trend Chart

The main chart is a line chart of Close Price with the 20 SMA and 50 SMA drawn on the same axes. I set the X-axis type to Continuous so the dates form an unbroken timeline, and I turned on the zoom slider so a period can be selected quickly. The legend is on to show which line is which.

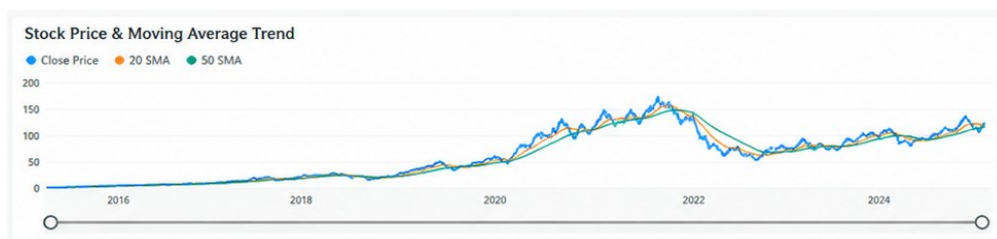


Figure 2: Stock price and moving average trend chart

- X-axis: Dim_Date[Date]
- Line values: Close Price, 20 SMA, 50 SMA
- X-axis type: Continuous
- Zoom slider: On

4. Trading Volume Chart

Below the price chart I added a second chart with Total Volume and the 20-day volume SMA. Days where the volume is more than 1.5 times the 20-day average are shown in red so the spikes are easy to see, and the normal days stay green. For this I made a colour measure and applied it to the columns with conditional formatting (fx).

```
Volume Color =  
IF(  
    [Total Volume] > [Volume 20 SMA] * 1.5,  
    "#FF4D4D",  
    "#26A69A"  
)
```

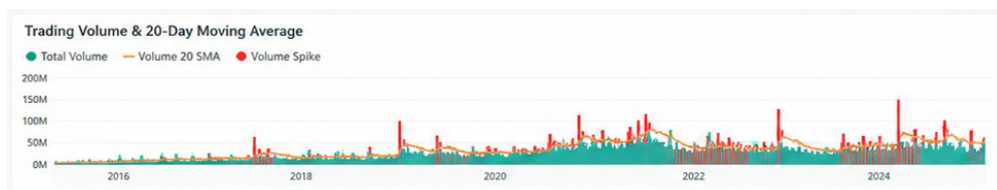


Figure 3: Trading volume with the 20-day SMA and volume spikes in red

5. Date Slicers and Metric Selector

I kept the date range slicer from Week 3 and set it to Between, so a start date and end date can be chosen. I then added a second slicer on Dim_Date[Date] and changed its type to Relative date. The screenshot shows it set to Last 1 month. The standard relative date slicer does not give separate buttons for 1 month, 3 months, 6 months, YTD and 1 year, so the interval has to be changed in the slicer itself.

For the Metric Selector I created a field parameter (Modeling > New parameter > Fields) with Close Price, High Price, Low Price and Open Price, and placed it in a slicer. Choosing a value in it changes the metric shown in the chart.

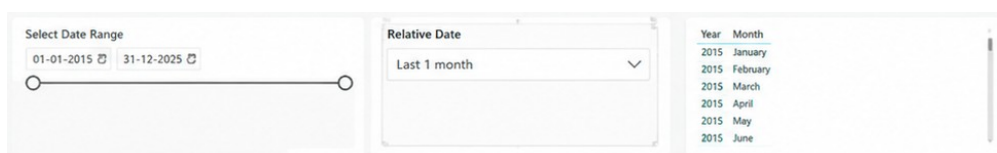


Figure 4: Date range slicer, relative date slicer and the year/month list

6. Tooltip Page

I created a second page named Stock_Tooltip that shows the details for the date the mouse is on. It has cards for Open, High, Low, Close, Total Volume and Daily Change %. To make it work as a tooltip, I turned on Allow as tooltip in the page information and changed the canvas type to Tooltip. Then on the main chart I went to Format visual > General > Tooltips, chose Report page and selected Stock_Tooltip.

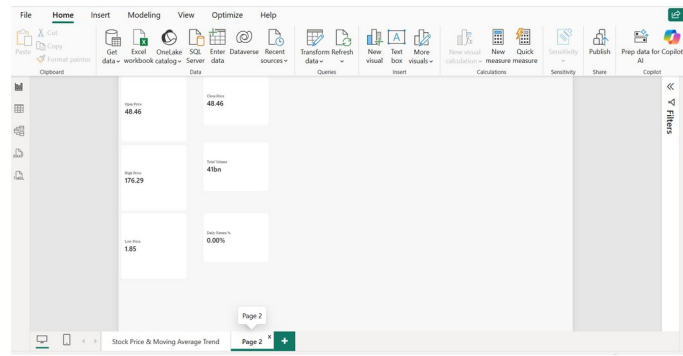


Figure 5: Stock_Tooltip page with the six cards

7. Dashboard Layout

I planned the layout so the two main charts stay large. The KPI cards and metric selector are at the top, the price chart is below them, then the volume chart, and the date controls and the year/month table are placed at the side and bottom. The tooltip page is a separate page, so it does not show on the dashboard.

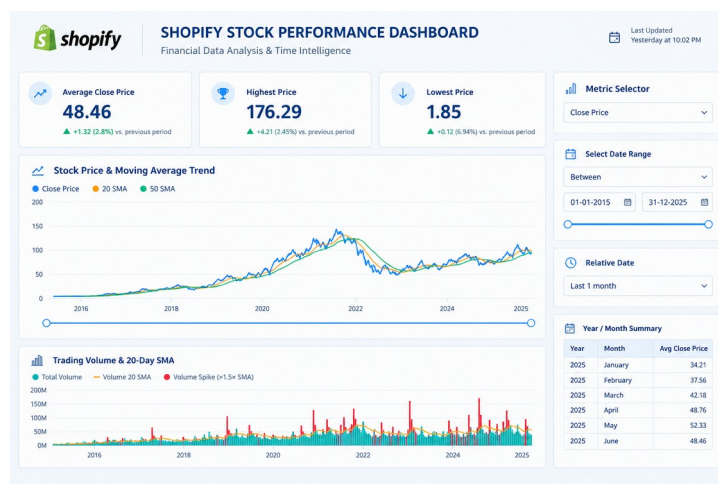


Figure 6: Design reference for the final layout (this is a mock-up, not a Power BI screenshot)

8. Checks Done

- 20 SMA and 50 SMA measures give values
- Crossover signal uses the correct measure names
- Price chart shows Close Price, 20 SMA and 50 SMA, with continuous axis and zoom slider
- Volume chart shows Total Volume and 20 SMA, with spike colours applied
- Date range slicer is on Between and the relative date slicer is working
- Metric selector has Open, High, Low and Close
- Stock_Tooltip page is set up and linked to the main chart
- Saved the Week 4 .pbix file

9. Submission

I upload the Week 4 PDF report and the dashboard PDF into the Week 4 folder in my GitHub repository (Shopify-Stock-Analysis), next to the Week 3 folder. The .pbix file is uploaded as well if GitHub accepts its size.